How to complete an outputs management plan

When applying for Wellcome funding you will need to provide a data sharing plan, or justify why a plan is not required. This page outlines some of the things you should consider as you complete your plan and implement your plan during your research.

This guidance should be read alongside our **policy on data sharing** and our **policy on intellectual property**.

In this guidance, data means quantitative or qualitative information generated over the course of research. This can take the form of datasets, code and software, images or data generated through analysis of samples or biological materials as well as intellectual property (IP) for example patents.

How we use your outputs management plan

When considering your application, we, the advisory committees and/or peer reviewers will assess your outputs management plan.

As part of the end-of-grant reporting process, we will consider the extent to which outputs have been managed and shared in line with our expectations – taking into account your outputs management plan and recognising that this may have been revised as your research progressed.

Choosing the right route: output sharing or IP and commercialisation

Outputs may be shared with end-users (openly or otherwise) or be made available commercially by licensing for a fee.

Your outputs management plan must set out which approach is most likely to maximise the adoption and use of the output by the wider research community and the resulting health benefit.

For example, if creating a new software tool, an open approach might be appropriate if others could make immediate and sustained use of it, (for example under a GNU General Public Licence or other licence approved by the **Open Source Initiative**).

However, a commercial approach might be better if you need further funding or a commercial partner to develop, market, distribute or support the ongoing use of the software.

You should also consider whether the output would have greater value to the research community if it was incorporated into an existing commercial product or an existing open resource, rather than making it available as a standalone product.

What to include in your plan

Your answers should be concise, precise and realistic. The detail should be proportionate to the complexity of the study, the types of data being managed, the anticipated long-term value of the data and the anticipated data security requirements.

Researchers must consider **Wellcome's Environmental Sustainability Funding policy** when developing their plan.

Where relevant, your plan must address the following:

1. **Data and software outputs**
   - **The data and software outputs your research will generate and/or re-use**
   - **The metadata and documentation that will accompany the outputs**
   - **When you intend to share your data and software**
   - **Where your data and software will be made available**
   - **How your data and software will be accessible to others**
   - **Whether limits to data and software sharing are required**
2. **Research materials**

   What materials your research will produce and how these will be made available.
3. **Resources and costs required**
   - **People and skills**
   - **Storage and computation**
   - **Access**
   - **Deposition and preservation of data, software and materials**

4. **Intellectual property**
    - **What IP your research will generate**
    - **How IP will be protected**
    - **How IP will be used to achieve health benefits**

1. Data and software outputs

The data and software outputs your research will generate and/or re-use

Consider and briefly describe:

- the types of data and software the proposed research will generate
- which data and software will have value to other research users and could be shared

We recognise that in some cases it may not be appropriate for researchers to share data and software outputs (for example, for ethical or commercial reasons). If you don't intend to share outputs, you must justify your reasons.

Software should be shared in a way that allows it to be used effectively, and you should provide appropriate and proportionate documentation for the user community.

We expect you to share null and negative findings and data, as well as data supporting new findings, where this may have value to the community. This helps to avoid unnecessary waste and duplication.

When existing data/resources are being re-used as part of the funding activity, you must consider:

- how existing data/resources will be accessed
- if there are any constraints on re-use of existing data
- how you will cite the original source, and credit the original creators appropriately

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The metadata and documentation that will accompany the outputs

Data should be shared in line with recognised data standards, where these exist, and in a way that maximises opportunities for data linkage and interoperability. **FAIRSharing** is one directory of available data standards.

You should:

- provide sufficient, high quality metadata to allow the dataset to be discovered, interpreted and used by others
- adopt agreed best practice standards for metadata provision, where these are in place, for example the **CDISC standards** for clinical research

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When you intend to share your data and software

You must specify the timescale for sharing datasets and software, using any recognised standards of good practice in your research field.

Researchers have the right to a reasonable (but not unlimited) period of exclusive use of the research data and software they produce.

As a minimum, you must make the data and software underpinning research articles available to other researchers **at the time of publication**, providing this is consistent with:

- any ethics approvals and consents that cover the data
- reasonable limitations required for the appropriate management and exploitation of IP

You must make sure that these articles include a statement explaining how other researchers can access the data, software or materials. This should be included in the Data Availability Statement field, or if this is not available, the statement should appear within the text of your manuscript.

Where research data relates to a public health emergency, quality-controlled data must be shared as rapidly and openly as possible. This is in line with the **joint statement on data sharing in public health emergencies** and **GLOPID-R principles for data sharing in public health emergencies**.

We encourage researchers to consider opportunities for timely and responsible pre-publication sharing of datasets and software. Where appropriate, you may use publication moratoria to enable pre-publication sharing with other researchers, while protecting your right to first publication.

Any restrictions on data and software use should be reasonable, transparent and in line with established best practice in the respective field.

Where your data and software will be made available

You must deposit data in recognised data repositories for particular data types where they exist, unless there's a compelling reason not to do so. The **FAIRSharing** and **Re3Data** sites provide lists of data resources, and Wellcome Open Research maintains a **curated list of approved repositories** suitable for Wellcome-funded research.

Where there is no recognised subject area repository available, researchers can use general community repositories and resources, such as **Dryad**, **FigShare**, the **Open Science Framework** or **Zenodo**.

If you intend to create a tailored database resource or to store data locally, you should ensure that you have the resources and systems in place to curate, secure and share the data in a way that maximises its value and guards against any associated risks.

You should consider how data held in this way can be effectively linked to and integrated with other datasets to enhance its value to users.

For software outputs, use a hosting solution that exposes them to the widest possible number of users. **GitHub** allows revision control and collaborative hosting of project code for software development. You must ensure that your code repository includes a licence, and has a persistent identifier (for example, a DOI). If you are using GitHub, you should use **Zenodo** to archive a repository on GitHub and issue a DOI for the archive.

How your data and software will be accessible to others

Your plan must set out clearly:

- how potential users will be able to discover, access and re-use data or software outputs
- any associated terms or conditions

Enabling discovery

Where a data or software resource is being developed as part of a funded activity, you should take reasonable steps to ensure that potential users are:

- made aware of its availability
- updated on significant revisions and releases

Your plan must outline your approach for maximising the discoverability of your data or software.

Researchers should use digital object identifiers (DOIs) or other persistent identifiers for their data and software outputs, to enable their re-use to be cited and tracked.

The **DataCite initiative** provides a key route through which DOIs are assigned to datasets. Many repositories assign DOIs on deposition.

Where appropriate, you may also publish an article describing dataset or software output to help users discover, access and reference the resource. You can use venues such as **Scientific Data**, **GigaScience** and **Wellcome Open Research**.

Access procedures for data

Where a managed data access process is required – for example, where a study involves identifiable data about research participants – the access mechanisms must be proportionate to the risks associated with the data. They must not unduly restrict or delay access.

You must describe any managed access procedures in your outputs management plan. It must be consistent, transparent and documented clearly on your study website.

Depending on the study, you may want to establish a graded access procedure where less sensitive data – for example, anonymised and aggregate data – is made readily available, and more sensitive datasets have a more stringent assessment.

Where a Data Access Committee is needed to assess data access requests, the committee should include individuals with appropriate expertise who are independent of the project.

Open software and database licences

If you're sharing your output through a repository, the terms by which you do so are likely to be set by the repository itself. If you're sharing directly with the research community, you must consider the most appropriate way to do so, for example by an appropriate open licence or public domain dedication.

For data, we recommend Creative Commons licences such as CC0 or CC BY. For software, the **Open Source Initiative** provides access to a range of open software licences, such as the GNU General Public Licence, Apache Licence, and the MIT Licence. Where possible, you should select one of these standard licences (rather than using a bespoke licence).

You must make sure it's clear which licence has been applied, so that users can see whether the data or software is accessible and on what terms.

Whether limits to data and software sharing are required

For some research, delays or limits on data sharing may be necessary to safeguard research participants or to ensure you can gain IP protection.

Restrictions should be minimised as much as possible and set out clearly in your outputs management plans, if required.

Safeguarding research participants and informed consent

For research involving human subjects, data must be managed and shared in a way that's fully consistent with the terms of the consent under which samples and data were provided by the research participants.

For prospective studies, consent procedures must include provision for data sharing in a way that maximises the value of the data for wider research use, while providing adequate safeguards for participants. For more information about consent for data sharing, go to the **UK**

Data Service. Procedures for data sharing must be set out clearly, and current and potential future risks explained to participants.

When designing studies, you must make sure that you protect the confidentiality and security of human subjects, including through appropriate anonymisation procedures and managed access processes as described on this page.

Clinical trials

For clinical trials you must mention specifically how you will share individual-level patient data. This includes:

- a plan to seek patient informed consent that allows data to be shared in the way outlined
- the level of identification risk and method of de-identification you will adopt
- the repository or platform you plan to use
- any managed access arrangements, such as a data access committee

Intellectual property (IP)

Delays or restrictions on data or software sharing may be appropriate to protect and use IP in line with our **policy on intellectual property and patenting**. If this applies, you should only share data or software when it no longer jeopardises your IP position or commercialisation plans.

Your proposed approach for identifying, protecting and using IP should be set out as described in the IP section of this guidance.

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2. Research materials

What materials your research will produce and how these will be made available

Your plan must identify any significant materials you expect to develop using Wellcome funding, which could be of potential value as a resource to other researchers.

Identify in your plan how the materials will be made available to potential users. For example, by:

- depositing in a recognised collection such as **ECACC**
- licensing to a reputable life science business partner who can handle advertising, manufacture, storage and distribution

If the material is highly specialised and the potential number of users is so small that commercial partners cannot be found, distributing samples yourself to other researchers who have asked for

them, may be an acceptable plan. However, where possible, you should find a more sustainable long-term solution that doesn't put an undue burden on you or your organisation.

When dealing with commercial entities, you must retain the right to produce the research materials yourself, and to license others to do so, if your chosen commercial partner is unable or unwilling to continue supplying them to the research community.

While your organisation may generate reasonable revenue from commercialising research materials, the primary driver should not be revenue generation. You should ensure that your research materials are made available to the wider research community and thereby advance the development of health benefits.

3. Resources and costs required

You should consider what resources you may need to deliver your plan and outline where dedicated resources are required. This includes people needed for the research as well as associated costs of the research.

Examples of resources you can ask for include:

People and skills

- support for one or more dedicated data manager or data scientist (full- or part-time)
- data and software management training for research or support staff that are needed to deliver the proposed research

We don't usually consider costs for occasional or routine support from organisational data managers or other support staff.

Storage and computation

- any dedicated hardware or software that is required to deliver your proposed research
- the cost of accessing a supercomputer or other shared facilities

We would usually expect costs associated with routine data storage to be met by the organisation. We will only consider storage costs associated with large or complex datasets which exceed standard organisational allowances.

Access

- the reasonable costs of operating an access committee or other data access mechanism over the lifetime of the award

- the costs of preparing and sharing data, software or materials with users (and whether cost-recovery mechanisms will be used)
- the costs of ingesting secondary data, code or materials from users
- costs associated with accessing data, software or materials from others researchers that you need to take forward your proposed research

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Deposition and preservation of data, software and materials

If no repository is suitable, we may consider ingestion costs for institutional repositories.

We don't usually consider estimated costs for curation and maintenance of data, code and materials that extend beyond the lifetime of the award. But we're willing to discuss how we can help support the long-term preservation of very high-value outputs on a case-by-case basis.

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4. Intellectual property

What IP your research will generate

Your plan should describe any significant IP that is likely to arise during your research. You should identify what processes you have in place to identify and capture this IP, as well as any unanticipated discoveries or inventions that result from your work.

How IP will be protected

You should describe if and how you will protect significant Wellcome-funded IP. For example, if you're registering a patent or design, you should briefly outline the territories in which you'll do this.

Publication of details relating to an invention can limit or entirely destroy the potential to patent and commercialise the invention in the future. If you think that patentable Wellcome-funded IP will arise (or when unanticipated IP has arisen), you should explain how you'll make sure that publications don't affect your ability to secure and make suitable use of patent protection to advance health benefits.

How IP will be used to achieve health benefits

Wellcome sees IP as a tool which can be used to advance health benefits. You should therefore focus on:

- the benefits your use of the IP will bring to the wider research community
- how this will benefit health

If your research output is particularly relevant to humanitarian or developing world issues, your plan should specifically address how:

- the output can best be made available for use internationally to address those issues
- your IP strategy will allow this

Where Wellcome-funded IP comprises a patentable invention, we expect in most cases that it will be protected by filing a patent application. This should be done at a time which maximises the prospects of achieving the desired health benefits, even if this requires a delay to publication. You should only publish details of a potentially patentable invention (without having first sought patent protection) where:

- A market assessment has been carried out and there is no credible prospect of a patent for that invention being commercialised now or in the near future.
- A deliberate decision not to patent the invention (and not to allow anyone else to patent) has been taken for policy reasons. Publication instead of patenting in this case should clearly benefit the wider research community and support the delivery of health benefits. Discuss this with your institution if you're unsure. Contact Wellcome for advice before publication if you're still unsure.

Revenue generation should only be a secondary consideration. The primary driver for any commercialisation must be to advance health benefit, even if your employer may generate revenue from commercialising Wellcome-funded IP.

Additional resources and examples

Data

- **FAIRsharing** – a curated and searchable portal of data standards, databases, and policies in the life sciences and other scientific disciplines.
- **Digital Curation Centre** – the UK's leading centre of expertise in data curation. The DCC provides a range of resources and training opportunities for the UK higher education sector, and has developed an online tool for developing data management plans in line with funder requirements.
- **Medical Research Council guidance and resources** – the MRC has developed detailed practical guidance for researchers on data sharing.
- **Cancer Research UK guidance** – CRUK guidance on creating a data management plan.
- **re3data.org** – a global registry of research data repositories across different academic disciplines.

- **UK Data Archive** – an internationally recognised repository of digital research data in the social sciences and humanities, with associated guidance and services for researchers.
- **Wellcome Sanger Institute open access science policy** – the Wellcome Sanger Institute has a policy setting out the principles that underlie data sharing at the Institute, with associated guidance for researchers.
- **TGHN Data sharing toolkit** – specifically for clinical data management and sharing advice and resources.

Software

- **Software Sustainability Institute** – the UK's leading source of expertise in research software sustainability. SSI offers training and advice targeting the specific concerns of research software, including a series of online guides on best practice in software development, licensing, repositories and using a software management plan.
- **Software Carpentry** – since 1998, Software Carpentry has been teaching basic software skills to researchers in science, engineering, and medicine. They run a worldwide training programme, and provide open access material for self-instruction.
- **eScience Lab** – eScience Lab host a suite of tools designed to support the creation of e-science laboratories. The tools have been adopted by a large variety of projects and institutions.
- **Open Source Initiative** – OSI provide the definition for open source software and maintain a list of licences that comply with that definition.

Materials

- **UK Health Security Agency - Culture Collections** - UK Health Security Agency is the custodian of four unique collections that consist of expertly preserved, authenticated cell lines and microbial strains of known provenance – namely the European Collection of Authenticated Cell Cultures (ECACC), the National Collection of Type Cultures (NCTC), the National Collection of Pathogenic Viruses (NCPV) and the National Collection of Pathogenic Fungi (NCPF).
- **Addgene** - Addgene is a global, nonprofit repository created to help researchers share and access plasmids.

Examples of output management plans

Read some real examples of what we consider to be good output management plans [PDF 172KB].

The examples are written by researchers working in a range of areas and at different career stages:

- Established researcher – neuroimaging data sharing
- Established researcher – genomic data sharing
- Clinician – controlled access to sensitive data
- PhD student – population modelling data sharing
- Early-career researcher – cardiology data and software sharing
- Mid-career humanities and social sciences researcher – sharing of transcripts and fieldnotes.

Contact us

Contact our information officers if you have a question about funding.

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